

The Three Brother's Mountains (although technically there are 4 brothers..)

Plot Numbers: 0114791, 0114792, 0114793, 0114794, 0114795, 0114796, 0114797 (2007), MP 281 (1976)



Dates	August 10 th – August 15 th
Location	The Three Brother's Mountains
Parking	Upper Parking Lot at Blackwell Peak
Campground	Kicking Horse Campground - Reservation: \$5/night booked from August 10-15
Estimated Total Hiking	- Upwards of ~60km! - AllTrails link: https://www.alltrails.com/trail/canada/british-columbia/kicking-horse-campground-via-heather-trail
Water Sources	- Small stream/ water source runs ~500m E/ before camp. Could dry up but should be present end of July. There should be a sign signifying the water source - Another small stream is present at camp, but it may dry up and it is next to the outhouse
Access to sites	- Access to plots from 2007: 0114791, 0114792, 0114793, 0114794, 0114795, 0114796, 0114797 from Heather Trail

	- Access to plots from 2019: 12-2000, 02-3088, 02-3085, 14-4577, 02-3089, 02-3090, 14-4384, 14-4383, 14-4325, 14-4326, 14-4323, 14-4324, 14-4361, 14-4382, 14-4385
Predicted Weather	As of July 8 th : N/A
Special Precautions	- Water sources may be limited but reviews say there is a small trickle through the creeks into end of August. - Bugs (always a concern) will be very abundant at camp - Reviews note there were many bear and deer sightings - The ridge can be quite windy and cold, should plan to bring layers for the sites in these areas
Packing List	- Will send out document later on
Common Species Hitlist	- Will send out a document later on

Flowers and meadows galore here!



Tentative hourly plans for the weekend:

Sunday August 10th (Travel and Hike in Day):

Total km of Hiking: ~13-15 km



1. **9:00am – 1:00pm:** Drive from UBC to Upper Parking Lot at Blackwell Peak, Manning Park
 - a. Pickup any last minute supplies, fill up water, fill up gas, ect. I will have picked up supplies the week before so there may be no need to stop!
2. **1:00pm – 6:00 pm:** Hike ~13km (436m elevation gain) to Kicking Horse camp
 - a. We will pass sites 0114791, 0114792, 0114793, 0114794, 0114795, 0114796, 0114797 on the way (and can scope out access) but will guaranteed not have time to survey
 - b. We may have time to pop up to the First Brother Mountain on the way to camp! It adds ~2km to total hiking. Plot 0114791 is along the trail on the way to summit. I doubt we'd have time to survey but we can always sus it out day-of.
3. **6:00pm – 7:30pm:** Set up camp and dinner
4. **7:30 – Sunset:** We might have some free time to check out the meadows/ walk around/ botanize if we want!
5. **10:00:** Back at camp and sleeping!

Monday August 11th Plan (Big Hike and 2-3 plot day):

Targeting sites: 00114791, 0114792, 0114793

Total km of Hiking: ~11km



1. **6:30 – 8:00:** Wake up and breakfast
2. **8:00 – 10:00:** Hike 4.63 km, 360m elevation, E on Heather Trail to plot 0114791 (right off of the trail)
3. **10:00 – 1:00:** Survey 0114791
4. **1:00 – 2:00:** Lunch
5. **2:00 – 2:30:** Ridge walk 600m W to 0114792
6. **2:30 – 5:30:** Resurvey 0114792
7. **(If time):** Ridge walk another 200m to 0114793
8. **(If time):** Resurvey 0114793
9. **6:00 – 7:30:** Walk 4.8km (41m elevation gain) back to camp
10. **7:30 – Sunset:** Dinner and some more free time to hang/ rest after all that hiking
11. **10:00:** Sleep!

Tuesday August 12th Plan (2-3 plot day, more hiking):

Targeting Sites: 0114793, 0114794, 0114796

Total km of Hiking: ~11km total



1. **6:30 – 8:00:** Wake up and breakfast
2. **8:00 – 10:00:** Hike 4.63 km, 360m elevation, E on Heather Trail to the ridge b/w 1st and 2nd brother. Walk ~600m to plot 0114793
3. **10:00 – 1:00:** Survey 0114793
4. **1:00 – 2:00:** Lunch
5. **2:00 – 2:30:** Ridge Walk another ~200m W to 0114794
6. **2:30 – 5:30:** Resurvey 0114794
7. **(If time):** Ridge Walk another 430m to 0114796
8. **(If time):** Resurvey 0114796
9. **6:00 – 7:30:** Walk 4.8km (41m elevation gain) back to camp
10. **7:30 – Sunset:** Dinner and some more free time to hang/ rest after all that hiking
11. **10:00:** Sleep!
12. **7:00 – Whenever:** Dinner somewhere or hike ~3km back to camp
13. **10:00 pm:** Sunset or back to camp early

Wednesday August 13th Plan (2-3 plot day, more hiking):

Targeting Sites: 0114796, 0114797, 0114795

Total km of Hiking: ~8km total

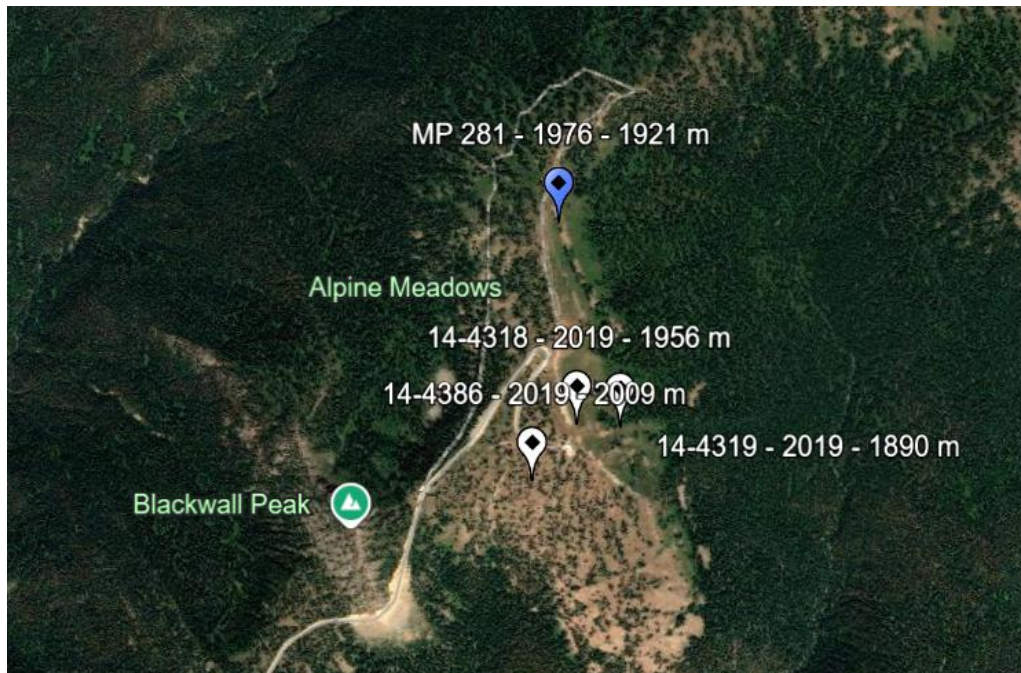
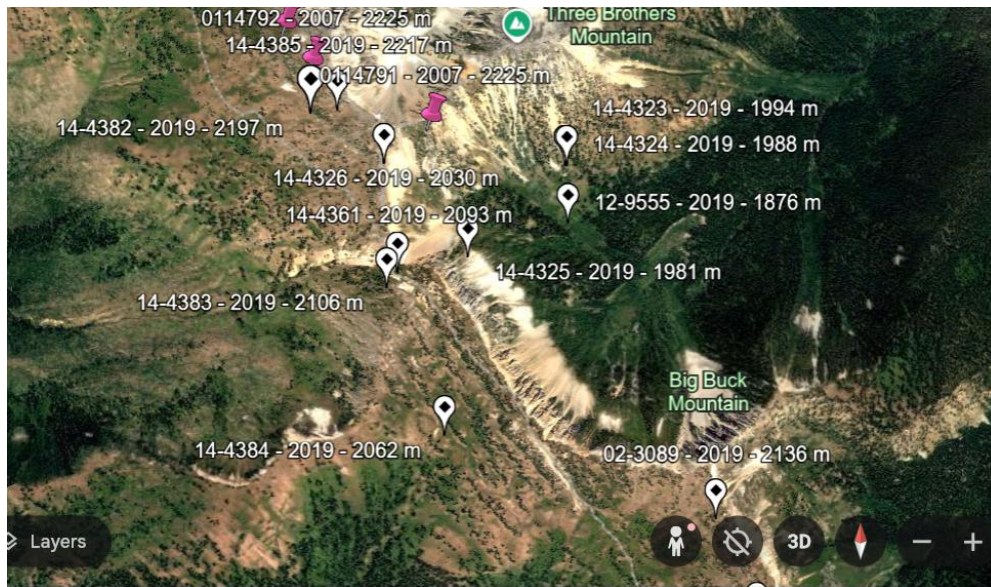


1. **6:30 – 8:00:** Wake up and breakfast
2. **8:00 – 10:00:** Hike 2.77 km, 168m elevation, E on Heather Trail. Walk ~300m to plot 0114796 to the ridge b/w 2nd brother and 3rd brother.
3. **10:00 – 1:00:** Survey 0114796
4. **1:00 – 2:00:** Lunch
5. **2:00 – 2:30:** Ridge Walk another ~300m W to 0114797
6. **2:30 – 5:30:** Resurvey 0114797
7. **(If time):** Ridge Walk another 200m to 0114795
8. **(If time):** Resurvey 0114795
9. **6:00 – 7:30:** Walk down to trail and 2.7 km back to camp
 - a. Or we could stay up on the ridge of the 3rd brother and eat dinner here
10. **7:30 – Sunset:** Dinner and some more free time to hang/ rest after all that hiking
11. **10:00:** Sleep!

Thursday August 14th Plan (1 plot day and hike down OR 2-3 plot day and stay another night):

Targeting Sites: 0114795, 14-4382,

Total km of Hiking: ~15km total



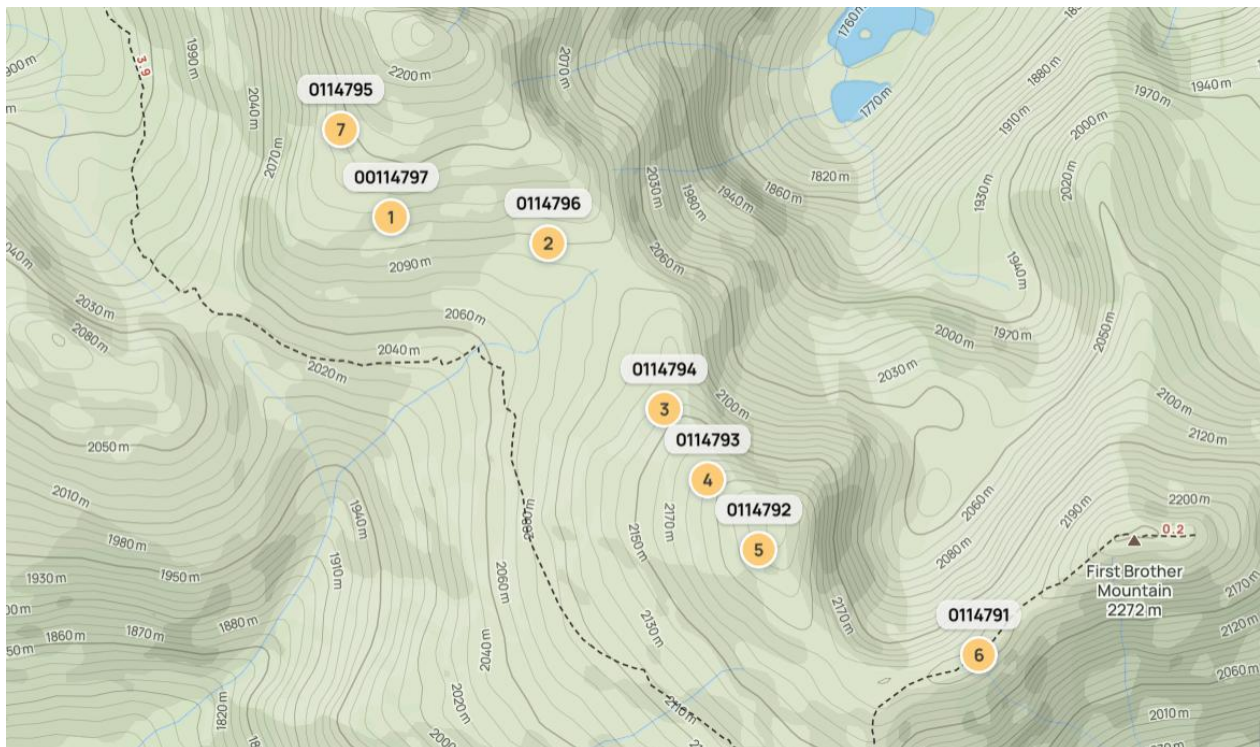
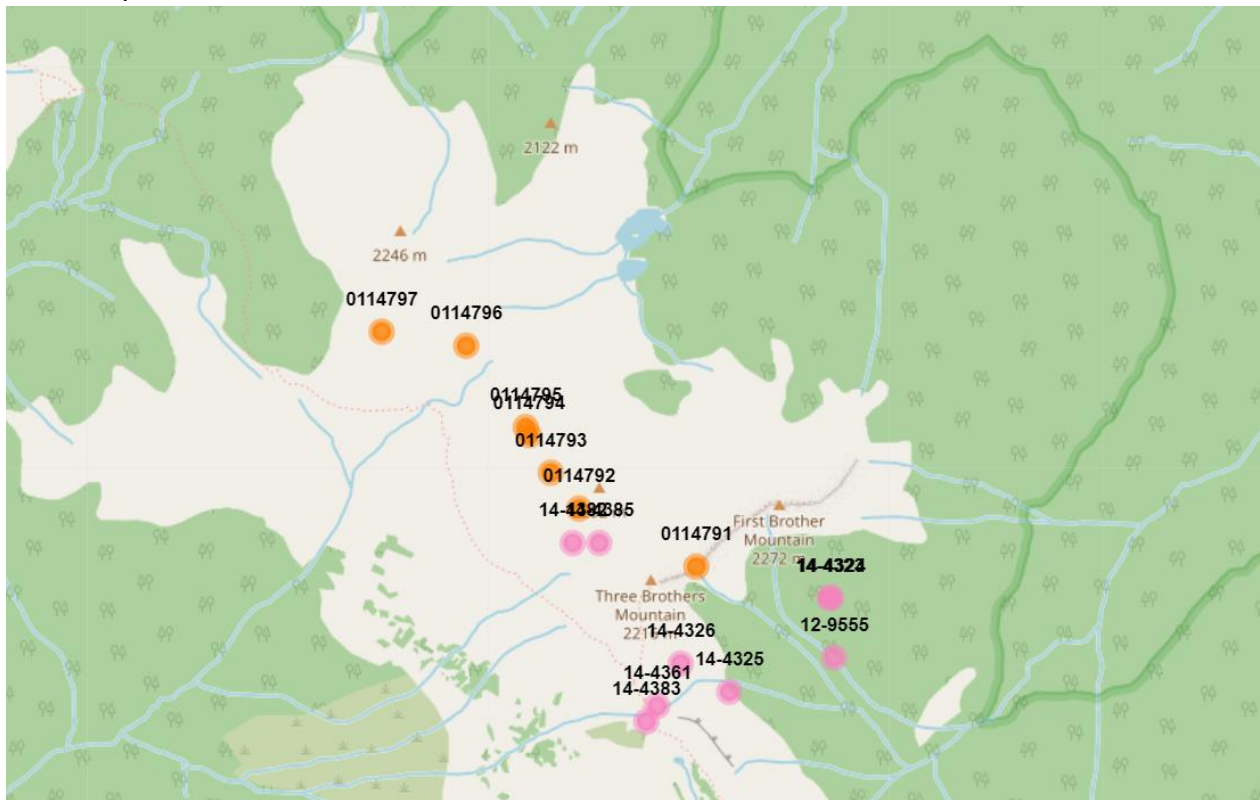
1. **6:30 – 8:00:** Wake up and breakfast
2. **8:00 – 10:00:** Hike 2 km, 150m elevation, E on Heather Trail. Walk ~450m upslope to plot 0114795 to the ridge b/w 2nd brother and 3rd brother.
3. **10:00 – 1:00:** Survey 0114796
4. **1:00 – 2:00:** Lunch
5. **2:00 – 6:00:** Hike 11km, 400m elevation gain, back to Blackwell Parking Lot
6. **(If time) Survey MP281 by parking**

7. 6:00 – 9:00: Drive back to Vancouver

Or stay another night and survey any of the 2019 sites: 14-4385, 14-4382, 14-4326, 14-4325

- We could hike to Buckhorn camp which is ~8.5 km, 300m elevation gain from Kicking horse camp and stay over here Thursday – Friday. I did not reserve Buckhorn, however, I doubt on a Thursday that it would be full. Some of the 2019 plots are closer to this camp

More Maps:



Plot Number	114791
Project ID	Alpine2007_BEC
Date Surveyed	2007-07-15
Access Via	Trail: E on Heather Trail ~4.6 km from camp
Location	Longitude: -120.7723007 Latitude: 49.1638298 Elevation: 2225 Location Notes: 3-Brothers Mtn, Manning Park, Cascades; first Brother Ridge
Site Information:	Stand age: NA Slope gradient: 45 Aspect: 140 Meso Slope Position: UP Surface shape: CV Zone and Subzone: ESSF mwp Site notes: NA
Plot Representation	Junicom - Arctuva - Dryaoct - Arencap
Species Recorded in original survey	Species list complete: FALSE PHLODIF, LUZUSPI, FESTBRA, JUNICOM, CAREALB, ARCTUVA, POLYPIL, RHODINT, POLEPUL, PINUALB, TONELYA, SABURUB, VACCSCO, POA_ALP, LUPILEP2, PELTIGER, PENSDAV, ACHIMIL, CAREPHA, ANTEROS, MICROCC, ERIGAU, EREMCAP, ABIELAS, THAMVER, DRYAHO, CERAPUR, POA_ARC, POTEGLU, SOLIMUL, FLAVNIV, COELACU, CARESCI, LESCRA, SELADEN, CASTRHE
Veg Notes	NA
Soil Notes	NA

Plot Number	114792
Project ID	Alpine2007_BEC
Date Surveyed	2007-07-14
Access Via	Trail: E on Heather Trail ~4.6 km from camp
Location	Longitude: -120.7787018 Latitude: 49.16588973 Elevation: 2225 Location Notes: 3 Brothers, Manning Park, Cascade Mtns; 2nd Brother
Site Information:	Stand age: NA Slope gradient: 10 Aspect: 225 Meso Slope Position: MD Surface shape: NA Zone and Subzone: ESSF mwp Site notes: slightly deeper depressions have Cassmer
Plot Representation	Vaccsco - Antelan - Juncpar - zonal? weak snowbed
Species Recorded in original survey	Species list complete: FALSE CASTPAR, CASSMER, HIERGRA, VEROCUS, JUNCPAR, ANTELAN, SIBBPRO, CETRISL, VACCSCO, PINUALB, CASTRHE, LUETPEC, POLYTRIC, CLADECM, SELADEN, LESCRA, PHLODIF, EREMCAP, PHYLEMP, LUPIARC
Veg Notes	NA
Soil Notes	NA

Plot Number	114793
Project ID	Alpine2007_BEC
Date Surveyed	2007-07-15
Access Via	Trail: E on Heather Trail ~5 km from camp
Location	Longitude: -120.7802963 Latitude: 49.16717147 Elevation: 2200 Location Notes: 3 Brothers, Manning Park, Cascade Mtns; 2nd Brother
Site Information:	Stand age: NA Slope gradient: 20 Aspect: 290 Meso Slope Position: CR Surface shape: CV Zone and Subzone: ESSF mwp Site notes: windswept but not insulated slopes wi this community
Plot Representation	Dryaoct - Saliniv - Carenar - Cetrniv windblown parkland ridge
Species Recorded in original survey	Species list complete: FALSE LUPILEP2, TONELYA, FLAVNIV, MICROCC, DRABA, DRYAHO, FLAVCUC, STELLON, SAXIAUS, SALINIV, BISTVIV, ALECOCH, SILEACA, THAMVER, TRISSPI, RHODINT, LUZUSPI, SELADEN, CAREPHA, CARENAR, CAREALB, ERIGAUR, PINUALB, POTEUNIF, ANTEALPI, LETHARIA, SABURUB, COELACU, PICEENE, FESTBRA
Veg Notes	NA
Soil Notes	NA

Plot Number	114794
Project ID	Alpine2007_BEC
Date Surveyed	2007-07-15
Access Via	Trail: E on Heather Trail ~5 km from camp
Location	Longitude: -120.7815018 Latitude: 49.16852951 Elevation: 2145 Location Notes: 3 Brothers, Manning Park, Cascade Mtns; 2nd Brother
Site Information:	Stand age: NA Slope gradient: 30 Aspect: 290 Meso Slope Position: MD Surface shape: CC Zone and Subzone: ESSF mwp Site notes: Recc plot
Plot Representation	Cassmer - (Vaccdel) - (Antelan) - snow bed in damp shallow hollow
Species Recorded in original survey	Species list complete: FALSE BRACHYT, ANELAN, CASTPAR, CARENIG, LUZUPIP, ARNILAT, PEDIBRA, VACCSCO, POTEFLA, SALINIV, RANUESC, PHYLEMP, VACCDEL, POHLIA, ABIELAS, LUETPEC, PICEENE, PELTIGER, CASSMER
Veg Notes	NA
Soil Notes	NA

Plot Number	114795
Project ID	Alpine2007_BEC
Date Surveyed	2007-07-15
Access Via	Trail: E on Heather Trail ~3 km from camp
Location	Longitude: -120.781601 Latitude: 49.16878128 Elevation: 2140 Location Notes: 3 Brothers, Manning Park, Cascade Mtns; 2nd Brother
Site Information:	Stand age: NA Slope gradient: 20 Aspect: 290 Meso Slope Position: MD Surface shape: CV Zone and Subzone: ESSF mwp Site notes: Recce plot
Plot Representation	Se - Phylgla - Empenig krummholz
Species Recorded in original survey	Species list complete: FALSE EMPENIG, CASSMER, LUPIARC, PHYLEMP, PINUALB, CARESPE, VEROCUS, POTEGLU, CETRISL, SIBBPRO, HIERGRA, CLADMIT, PHYLGLA, PICEENE, ARNILAT, EREMCAP, CLADECM, ABIELAS, ANTEALPI, ANTELAN, DICRFUS, SALICAS
Veg Notes	NA
Soil Notes	hasty soil

Plot Number	114796
Project ID	Alpine2007_BEC
Date Surveyed	2007-07-15
Access Via	Trail: E on Heather Trail ~3 km from camp
Location	Longitude: -120.7848969 Latitude: 49.17169952 Elevation: 2100 Location Notes: 3 Brothers, Manning Park, Cascades - 3rd Brother
Site Information:	Stand age: NA Slope gradient: 15 Aspect: 190 Meso Slope Position: LW Surface shape: NA Zone and Subzone: ESSF mwp Site notes: distinctly rototilled by rodents. Abundant bare earth and tunnels. Common type though with some variability. See notebook 2007. Recce Plot no soils
Plot Representation	Rodent turbated meadow Anemocc - Luzupip - Vahlatr - Carespe - Valesit
Species Recorded in original survey	Species list complete: FALSE PHLODIF, DRABAU, EPILANA, CASTPAR, VALESIT, VACCSCO, VEROCUS, PULSOCC, CARESPE, RANUESC, JUNC PAR, VERA VIR, EPILCIL, CARENIG, POA_CUS2, FESTSAX, SENEINT, LUPIARC, ARNIRYD, POTEFLA, LUZUPIP, SENETRI, EREMCAP, CLAYLAN
Veg Notes	Too early for veg. Still flushing missed species and cover will be higher later in season.
Soil Notes	NA

Plot Number	114796
Project ID	Alpine2007_BEC
Date Surveyed	2007-07-15
Access Via	Trail: E on Heather Trail ~3 km from camp
Location	Longitude: -120.7894974 Latitude: 49.17219924 Elevation: 2100 Location Notes: 3 Brothers, Manning Park, Cascades - 3rd Brother
Site Information:	Stand age: NA Slope gradient: 30 Aspect: 180 Meso Slope Position: MD Surface shape: NA Zone and Subzone: ESSF mwp Site notes: recce plot
Plot Representation	Fest sax?? - Arenca - Arnica rodent turbated dry slope
Species Recorded in original survey	Species list complete: FALSE ERIGAU, PEDILAB, SELADEN, JUNCAP, SILEPAR, AGOSAU, ANTEALPI, PENSPO, ARNIPAR, LUPIARC, ANELAN, ARNIRYD, DELPMEN, EREMCAP, ERIGPER, POLYPIL, PULSOCC, ACHIMIL, LUZUSPI, CASTRHE, FESTVIR, PHLEALP, VEROCUS, CAREROS, CAREPHA, SENEINT, EPILANA, CLAYLAN, POLYGONU, PHLODIF, MICROCC, POTEGLU, SEDULAN
Veg Notes	NA
Soil Notes	hasty pit

Plot Number	MP 281
Project ID	BEC-Vancouver
Date Surveyed	1976-09-27
Access Via	Trail: Lone Man Ridge Parking: Three brothers trailhead parking
Location	Longitude: -120.7583008 Latitude: 49.10832977 Elevation: 1921 Location Notes: Manning Park
Site Information:	Stand age: NA Slope gradient: 25 Aspect: 205 Meso Slope Position: UP Surface shape: CV Site notes: Project ID: EP 793; ESSFdk to ESSFvc. Tentative SNR keyed out post hoc (KS 2009) - too rich?
Plot Representation	NA
Species Recorded in original survey	Species list complete: FALSAE PELTCAN, PEDIRAC, VALESIT, VIOLORB, VACCSCO, CALARUB, NIPHCAN, FESTOCC, EUCEENG, ARNILAT, HIERACIU, POHLNUT, ACHIMIL, ORTHSEC, LUPIARC, PEDIBRA, SELAWAL, FRAGVES, PAXIMYR, CASTILLE, LUZUHIT, POLYPIL, DICRPAI, CLADONIA, BRACHYT, LONIUTA, ABIELAS, PINUALB, PICEENE
Veg Notes	CLADONI is Cladonia apodocarpa which is not in the provincial species list.
Soil Notes	NA

Plot Number	02-3085
Project ID	BC Parks Manning TEM
Date Surveyed	2019-10-29
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7638931 Latitude: 49.14056015 Elevation: 2044 Location Notes: 3 Brothers, Manning Park
Site Information:	Stand age: NA Slope gradient: 9 Aspect: 228 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mwp Site notes: draft BEC mapping - adjacent to ESSFdc2. Low shrubs w/ occasional trees (well developed structure but under 10 m). Rocks nearby but not in plot. Mineral largely covered by organic duff.
Plot Representation	Bl - Sx - Vacc cesp - Ante lan parkland
Species Recorded in original survey	Species list complete: TRUE ANTELAN, ERIGGLA, SILEPAR, PINUALB, LUZUPAR, LUPIARC, HIERGRA, PHLODIF, JUNCPAR, LUETPEC, VACCCEAE, LUPILEP2, VACCSCO, PULSOCC, LICHEN, PEDIBRA, ABIELAS, POTEFLA, PICEENE, FESTVIR, VACCMEM, EREMCAP, VACCDEL, VAHLATR
Veg Notes	Crustose lichens. A3 trees are technically the vets (A1) but to stay consistent w/ SS 3b, interpret accordingly.
Soil Notes	NA

Plot Number	02-3087
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-14
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.752121 Latitude: 49.1199913 Elevation: 1874 Location Notes: 3 Brothers, Manning Park
Site Information:	Stand age: NA Slope gradient: 14 Aspect: 258 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mww Site notes: NA
Plot Representation	Bl - Sx - Vaccsco - Luzula parv
Species Recorded in original survey	Species list complete: TRUE POLYJUN, ARNILAT, LUPIARC, VACCDEL, HYLOTRI, VACCSCO, ABIELAS, VIOLA, LUZUPAR, PINUCON2, VALESIT, PICEENE, ORTHSEC, VAHLATR
Veg Notes	Pl dead in A canopy
Soil Notes	c.f. lith - soft grainy rock dominates, same as HK2.

Plot Number	02-3088
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-14
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7645264 Latitude: 49.13434982 Elevation: 1928 Location Notes: 3 Brothers, Manning Park
Site Information:	Stand age: NA Slope gradient: 24 Aspect: 164 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mww Site notes: Benches while walking downslope. Midslope but has components of upper slope.
Plot Representation	Bl - Sx forb woodland
Species Recorded in original survey	Species list complete: TRUE THALOCC, PEDIRAC, ARNILAT, PICEENE, ELYMGLA, ABIELAS, POTEGLU, ARNICOR, LUZUPAR, RHODALB, BRACHYT, VALESIT, OZOMTRI, VIOLA, TROLALB, ACHIMIL2, CAREROS, CASTMIN, VACCSCO, LUPIARC, ERIGGLA, VERAVER, VACCMEM, HERAMAX, SENETRI, PARNFIM, PULSOCC, LONIINV, POTEFLA, PEDIBRA
Veg Notes	rich forbs, few shrubs
Soil Notes	NA

Plot Number	02-3089
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-15
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7567673 Latitude: 49.14989852 Elevation: 2136 Location Notes: 3 Brothers, Manning Park
Site Information:	Stand age: NA Slope gradient: 12 Aspect: 20 Meso Slope Position: UP Surface shape: ST Zone and Subzone: ESSF mwp Site notes: Submesic. Old in parkland so structural stage 3b with trees not over 10m. Wind, cold. We think Ah but not carpets of heather.
Plot Representation	Cliff edge/ Boundary of ESSFmwp and ESSFdcp. Bl - Cassiope - Phyllo - Luzupar
Species Recorded in original survey	Species list complete: TRUE LUPILEP2, PHLEALP, CAREMIP, TRISSPI, VEROCUS, LUZUPAR, FESTVIR, EREMCAP, JUNCPAR, PHYLEMP, ERIGGLA, LUETPEC, POA_CUS, PENSDAV, ERIGAU, LUPIARC, JUNICOM, VACCSCO, VAHLATR, LUZUSPI, CARESPE, CASTMIN, ABIELAS, HIERGRA, CAREPHA, PINUALB, ANELAN, CASSMER, CASTPAR, ARNILAT
Veg Notes	NA
Soil Notes	same hard fine grained bedrock/coarse frag lit.

Plot Number	02-3090
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-15
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7544632 Latitude: 49.1461296 Elevation: 2119 Location Notes: Polygon 1701, below crest. 3 Brothers, Manning Provincial Park
Site Information:	Stand age: NA Slope gradient: 10 Aspect: 144 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF dcp Site notes: zonal-ish plot. Mesoslope MD(UP). Parkland with old trees not over 10m. Rodent tunnels and dens throughout the site without trees. NR between C-D, exposed mineral soil from rodents
Plot Representation	Sx - Bl - festvir - lupiarc
Species Recorded in original survey	Species list complete: TRUE PICEENE, AGOSAUR, POTEFLA, PULSOCC, HIERGRA, PHLEALP, VAHLATR, VALESIT, VEROCUS, EREMCAP, LUZUPAR, VACCSCO, CASTPAR, LUETPEC, CARESPE, JUNCPAR, ABIELAS, POA_CUS, ARNILAT, PEDIBRA, PINUALB, FESTVIR, TRISSPI, ANTELAN, ERIGGLA, PHYLEMP, LUPIARC, SENEINT
Veg Notes	dense luzupar under trees
Soil Notes	bedrock/c.f. lith - same fine textured hard. Organic material mixed into upper Bf layer where CF less dense

Plot Number	12-2000
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-15
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7637634 Latitude: 49.13544082 Elevation: 2119 Location Notes: 10 m off Heather Trail, Manning P.P
Site Information:	Stand age: 102 Slope gradient: 27 Aspect: 144 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mww Site notes: SNR (C)D. woodland, no rodent activity, grouse presence
Plot Representation	Sx Bl veravir valesit woodland
Species Recorded in original survey	Species list complete: TRUE PECTBRE, PICEENE, PODAHUM, THALOCC, ERIGGLA, VACCSCO, LUZUPAR, VAHLATR, PULSOCC, VACCMEM, TROLALB, ELYMGLA, VALESIT, VIOLA, EPILGLA, BRACHYT, ROELROE, ABIELAS, LUPIARC, PEDIBRA, VERAVER, POTEFLA, PEDIRAC, ARNILAT
Veg Notes	NA
Soil Notes	NA

Plot Number	14-4577
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-29
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7613602 Latitude: 49.14511871 Elevation: 2119 Location Notes: Heather Trail near Big Buck Mountain, Manning Provincial Park
Site Information:	Stand age: NA Slope gradient: 34 Aspect: 290 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mwp Site notes: Lots of rodent digging
Plot Representation	Fescue - Valerian - Lupine meadow with lots of rodent activity
Species Recorded in original survey	Species list complete: TRUE POA_CUS, POTEDRU, PULSOCC, FESTVIR, AGOSGLA, PHLEALP, TRISSPI, LUPIARC, VALESIT, THALOCC, ARNILAT, POLYDOU, EUCEENG, POTEFLA, CARESPE, SILEPAR, SENEINT
Veg Notes	NA
Soil Notes	NA

Plot Number	14-4384
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-29
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7717896 Latitude: 49.15280151 Elevation: 2062 Location Notes: Heather Trail near Big Buck, Manning Provincial Park
Site Information:	Stand age: NA Slope gradient: 20 Aspect: 210 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mwp Site notes: SMR 3(4). Treed area interspersed by open meadows. Rodent digging.
Plot Representation	Sx - Bl - valesit - senetri - arnica moderately sloped parkland
Species Recorded in original survey	Species list complete: TRUE VACCSCO, LUPIARC, ERIGGLA, TROLALB, SENETRI, AULAPAL, ARNILAT, ABIELAS, PEDIGRO, FESTVIR, PECTBRE, THALOCC, LUZUPAR, HIERUMB, VERAVER, EUCEENG, PEDIBRA, GALITRE, ARNICOR, CASTMIN, RANUESC, CASTPAR, LEPTPYR, VAHLATR, VALESIT, POTEFLA, PODAHUM, PICEENE, CARESPE, CALTLEP
Veg Notes	NA
Soil Notes	NA

Plot Number	14-4383
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-29
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7749863 Latitude: 49.15822982 Elevation: 2106 Location Notes: Trail between Big Buck and First Brother Mountain, Heather Trail, Manning PP
Site Information:	Stand age: NA Slope gradient: 52 Aspect: 328 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mwp Site notes: Lots of rodent digging. Note SS3b because <10% tree cover, though mature Bl present in canopy. Lots of structure for Parkland. "Tree island"
Plot Representation	Sx - Bl - Pa - Luzupar - Vaccsco - Lupiarc
Species Recorded in original survey	Species list complete: TRUE VACCSCO, POA_CUS, CLADONIA, CARESPE, POLYJUN, LUZUPAR, VAHLATR, PHYLEMP, PHLEALP, LUETPEC, ARNILAT, FESTVIR, CASSMER, VEROCUS, ERIGGLA, HIERUMB, ABIELAS, PICEENE, BRACHYT, VALESIT, CETRARIA, ANTELAN, PINUALB, TRISSPI, CASTPAR, LUPIARC
Veg Notes	NA
Soil Notes	NA

Plot Number	14-4325
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-29
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7705002 Latitude: 49.15930175 Elevation: 1981 Location Notes: First Brother, 3 Brothers Ridge, Manning PP
Site Information:	Stand age: NA Slope gradient: 30 Aspect: 100 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mwp Site notes: SCR = D
Plot Representation	LaBl - Vaccdel - Phylemp late snowbed
Species Recorded in original survey	Species list complete: TRUE CARENIG, LUZUPAR, CASSMER, VERAVER, VACCDEL, ARNILAT, ABIELAS, CLADECM, VAHLATR, PICEENE, PEDIBRA, PHYLEMP, BRACHYT, VALESIT, LARILYA, LUETPEC, VACCSCO, BARBILOP, ROELROE
Veg Notes	NA
Soil Notes	NA

Plot Number	12-9555
Project ID	BC Parks Manning TEM
Date Surveyed	2019-10-31
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.76474 Latitude: 49.16053009 Elevation: 1876 Location Notes: First Brother, 3 Brothers ridge, Manning PP
Site Information:	Stand age: NA Slope gradient: 20 Aspect: 155 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mww Site notes: NA
Plot Representation	Aster engelmannii - thalocc - grass avalanche slope
Species Recorded in original survey	Species list complete: TRUE THALOCC, PEDIBRA, ARNIPAR, FESTVIR, CHAMANG, VALESIT, ARNILAT, AQUIFOR, BROMCAR, ABIELAS, SILEPAR, HACKMIC, PULSOCC, CIRSEDU, ANGEARG, SYMPFOL, AGOSAUR, HERAMAX, MITENUD, LUPIARC, VERA VIR, SEDULAN, ELYMGLA, EUCEENG, ACHIMIL
Veg Notes	NA
Soil Notes	Ahb horizons - buried organic inputs - continual accumulation process

Plot Number	14-4324
Project ID	BC Parks Manning TEM
Date Surveyed	2019-08-29
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7649002 Latitude: 49.16270065 Elevation: 1988 Location Notes: First Brother, 3 Brothers Ridge, Manning PP
Site Information:	Stand age: NA Slope gradient: 35 Aspect: 140 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mww Site notes: Structural stage 6/7. Adjacent to rock talus and avalanche. Hasty plot.
Plot Representation	Bl - Luzupar - valesit mature forest patch of upper woodland/parkland
Species Recorded in original survey	Species list complete: TRUE SORBSIT, VIOLA, VACCSCO, ORTHSEC, LUZUPAR, RHODALB, PHYLEMP, PECTBRE, BRACHYT, RUBUPED, ABIELAS, ARNILAT, VACCDEL, VALESIT, STRELAN, VERA VIR, PICEENE, VIOLORB, VACCMEM
Veg Notes	Hasty plot
Soil Notes	No pit dug

Plot Number	14-4323
Project ID	BC Parks Manning TEM
Date Surveyed	2019-11-29
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7649994 Latitude: 49.16270065 Elevation: 1994 Location Notes: First Brother, 3 Brothers Ridge, Manning Provincial Park
Site Information:	Stand age: NA Slope gradient: 75 Aspect: 215 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mwp Site notes: NA
Plot Representation	Pa - Junicom - paximyr - phlodif mixed grass parkland on steep warm aspect Cv
Species Recorded in original survey	Species list complete: TRUE ANTEALPI, VACCSCO, KOELMAC, EREMCAP, POA_SEC2, SILEPAR, AQUIFOR, ANTEUMB, ARNIPAR, CAREPHA, PENSPRO, SOLIMUL, THALOCC, ERIOUMB, VAHLATR, ABIELAS, OSMORHIZ, ARCTUVA, PINUALB, PHLODIF, JUNICOM, AGOSAUR, ARNILAT, ACHIMIL, SEDULAN, FESTVIR, CASTMIN, CHAMANG, CALARUB, FRAGVIR, PAXIMYR
Veg Notes	some dead, some healthy whitebark pine
Soil Notes	Bedrock outcrops adjacent

Plot Number	14-4326
Project ID	BC Parks Manning TEM
Date Surveyed	2019-11-29
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7731018 Latitude: 49.1603012 Elevation: 2030 Location Notes: First Brother on three brothers ridge, Manning PP
Site Information:	Stand age: NA Slope gradient: 65 Aspect: 140 Meso Slope Position: MD Surface shape: ST Zone and Subzone: ESSF mwp Site notes: SCR C+. Hasty plot
Plot Representation	Anemocc - Festvir - valesit on steep colluvial snow accumulating slope, ravelling
Species Recorded in original survey	Species list complete: TRUE VERAVIR, ARNIPAR, VEROCUS, POTEFLA, CASTMIN, POTENAN, CARESPE, VACCDEL, AGOSAUR, VACCSCO, ACHIMIL, VALESIT, LUZUPAR, ANTELAN, PENSPRO, PHLODIF, ERYTGRA, PHYLEMP, ERIGGLA, PHLEALP, FESTVIR, PULSOCC, POLYPIL
Veg Notes	some dead, some healthy whitebark pine
Soil Notes	Bedrock outcrops adjacent

Plot Number	14-4385
Project ID	BC Parks Manning TEM
Date Surveyed	2019-11-29
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7775726 Latitude: 49.16463088 Elevation: 2217 Location Notes: Crest of First Brother Mountain, Manning Provincial Park
Site Information:	Stand age: NA Slope gradient: 35 Aspect: 270 Meso Slope Position: CR Surface shape: CV Zone and Subzone: ESSF mwp Site notes: SMR 0(1), SNR C-D. Lots of dead cushion plants
Plot Representation	NA
Species Recorded in original survey	Species list complete: TRUE DRYADRU, SABURUB, THAMVER, POLYJUN, SEDULAN, SELASCO, POTEGLU, MICRINT, FESTBRA, PINUALB, CAREPHA, JUNICOM, CAREFIL, SOLIMUL, ANTEROS, SAXIAUS, LUZUSPI, STELLON, CLADONIA, TRISSPI, PENSPRO, ERIGAUR, ARCTUVA, ABIELAS
Veg Notes	NA
Soil Notes	NA

Plot Number	14-4382
Project ID	BC Parks Manning TEM
Date Surveyed	2019-11-28
Access Via	Trail: E on Heather Trail
Location	Longitude: -120.7790527 Latitude: 49.1646614 Elevation: 2197 Location Notes: Trail to first brother, Manning Provincial Park
Site Information:	Stand age: NA Slope gradient: 46 Aspect: 245 Meso Slope Position: UP Surface shape: ST Zone and Subzone: ESSF mwp Site notes: SMR 2(3). Trees ~ 10m tall, fairly stunted. Lots of rodent digging
Plot Representation	Bl arnilat - phlodiff forb-rich meadow with lots of rodent digging
Species Recorded in original survey	Species list complete: TRUE PINUALB, SEDULAN, ACHIMIL2, ARNILAT, PHLEALP, CIRSEDU, PENSPRO, POTEGLU, SIBBPRO, ARNIPAR, POLYDOU, HERAMAX, SILEPAR, EREMCAP, LUZUSPI, TRISSPI, VERAVER, LUPILEP2, ERIGGLA, PHLODIF, SENETRI, BOECLEM, POA_CUS, LUPIARC, ABIELAS, POA_GLA, LUZUPAR, EPILANA, CASTMIN, SOLIMUL
Veg Notes	ground too turbated for mosses and lichens
Soil Notes	NA